RK Health – Smart Patient Appointment & Medication Reminder System

Project Description

RK Health is an AI-powered healthcare management system designed to simplify patient healthcare services through appointment scheduling, medication reminders, AI-generated medical summaries, and health report management. The application enables patients to efficiently manage doctor appointments, medication schedules, and personal health records through a secure and user-friendly web interface.

The system integrates **Google Apps Script** as the backend, **Google Sheets** as the cloud database, **Groq AI API** for generating intelligent visit summaries, **Google Calendar API** for appointment scheduling, and **Twilio SMS API** for medication reminder notifications. The frontend is developed using **HTML, CSS, and JavaScript**, providing a responsive and intuitive user experience.

The application is deployed using **GitHub Pages**, while backend services are hosted through **Google Apps Script Web App**, enabling cloud-based data storage and real-time access without requiring a dedicated server.

RK Health provides an affordable, scalable, and intelligent healthcare solution that assists patients in organizing healthcare activities, reducing missed appointments, improving medication adherence, and maintaining digital health records.

Scenarios

Scenario 1 – Patient Appointment Booking

A patient logs into RK Health, schedules a doctor's appointment, selects the preferred date and time, and the system automatically stores the appointment in Google Sheets while creating an event in Google Calendar.

Scenario 2 – Medication Reminder

The patient adds medicine details including dosage and reminder time. RK Health stores the medication schedule and automatically sends reminder notifications using Twilio SMS to ensure medicines are taken on time.

Scenario 3 – AI Health Summary Generation

After completing a consultation, the patient uploads or enters visit details. RK Health uses the Groq AI API to generate a simplified medical visit summary that helps patients easily understand prescriptions, diagnoses, and follow-up instructions.

Technical Architecture:

```
                    +----------------------+
                    |      Patient         |
                    +----------------------+
                               |
                               |
                    HTML • CSS • JavaScript
                               |
                               v
               +--------------------------------+
               |     RK Health Frontend          |
               | (Login, Dashboard, Reports)    |
               +--------------------------------+
                               |
                    API Requests (JSON)
                               |
                               v
             +-----------------------------------+
             | Google Apps Script Backend (API)  |
             +-----------------------------------+
                /          |             |            \
               /           |             |             \
              v            v             v              v
   Google Sheets      Groq AI API   Google Calendar   Twilio SMS
   Cloud Database      AI Summary      Scheduling      Notifications
              \            |             |             /
               \           |             |            /
           +-------------------------------------+
           |     Patient Healthcare Services     |
           +-------------------------------------+
```

Description: RK Health follows a **cloud-based three-tier architecture** consisting of the Presentation Layer (Frontend), Business Logic Layer (Backend), and Data Layer (Cloud Database). The frontend is developed using **HTML, CSS, and JavaScript**, providing a responsive interface for patients to access healthcare services. The backend is implemented using **Google Apps Script**, which processes API requests, validates user data, communicates with cloud services, and performs business logic.

(Note: If you are using database in your project definitely you need to add ER Diagram along with description)

Pre-requisites:

- **HTML, CSS, and JavaScript Skills:** https://www.w3schools.com/html/
- **Google Apps Script Knowledge:** https://developers.google.com/apps-script

- **Google Sheets API Familiarity:** https://developers.google.com/sheets/api

- **Groq AI API Integration:** https://console.groq.com/docs

- **Google Calendar API:** https://developers.google.com/calendar

- **Twilio SMS API:** https://www.twilio.com/docs

- **Git & GitHub Version Control:** https://git-scm.com/doc

- **GitHub Pages Deployment:** https://pages.github.com/

- **Visual Studio Code IDE:** https://code.visualstudio.com/docs

- **Basic REST API & JSON Knowledge:** https://developer.mozilla.org/en-US/docs/Learn_web_development/Extensions/Server-side/First_steps/Client-Server_overview

Project Workflow:

- **Milestone 1: Project Planning & UI Design:**
- **Defined project objectives and requirements.**
- **Designed UI wireframes for Login, Dashboard, Appointments, Medicines, and Reports.**
- **Selected the technology stack (HTML, CSS, JavaScript, Google Apps Script, Google Sheets).**
- **Deliverables**
- **Project proposal**
- **UI design**
- **Technology selection**
-

Milestone 2: Backend Development & Database

Activities

- Developed backend using Google Apps Script.
- Configured Google Sheets as the cloud database.
- Implemented CRUD operations for appointments, medicines, and health records.

Deliverables

- Backend APIs

- Database integration
- CRUD functionality

Milestone 3: AI & External Service Integration

Activities

- Integrated Groq AI for healthcare summaries.
- Connected Google Calendar for appointment scheduling.
- Added Twilio SMS for medication reminders.

Deliverables

- AI Summary module
- Calendar integration
- SMS reminder system

Milestone 4: Testing & Deployment

Activities

- Tested all project modules.
- Fixed bugs and optimized performance.
- Deployed the application using GitHub Pages and Google Apps Script.

Deliverables

- Tested application
- Live deployment
- Final project documentation

Milestone 1: Project Setup and Backend Configuration

In this milestone, we focused on setting up the development environment and configuring the backend services required for RK Health. The frontend was developed using HTML, CSS, and JavaScript, while the backend was implemented using Google Apps Script. Google Sheets was configured as the cloud database to store patient records, appointments, medicines, and AI summaries.

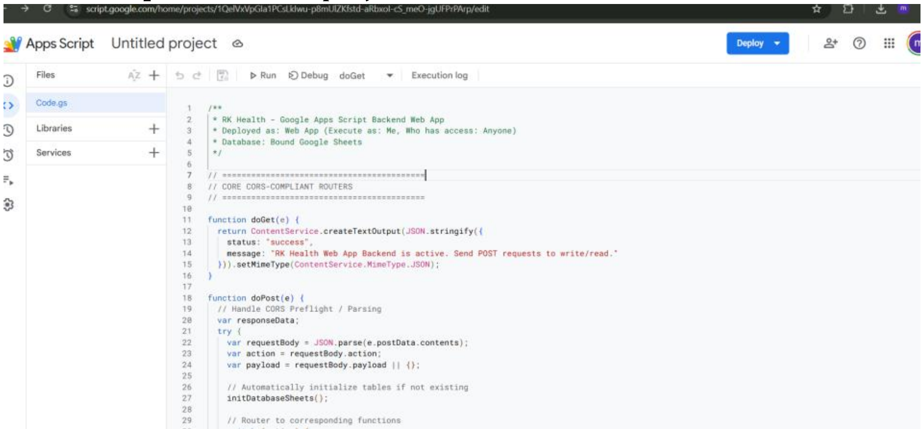
Activity 1.1: Create Google Apps Script Project

Before developing the backend, a Google Apps Script project was created.

Steps

- **Step 1** – Open **Google Apps Script** (https://script.google.com).
- **Step 2** – Create a new Apps Script project named **RK Health Backend**.

- **Step 3** – Replace the default code with the project backend (Code.gs).
- **Step 4** – Save the project.



Activity 1.2: Configure Google Sheets Database

Google Sheets was used as the cloud database.

Steps

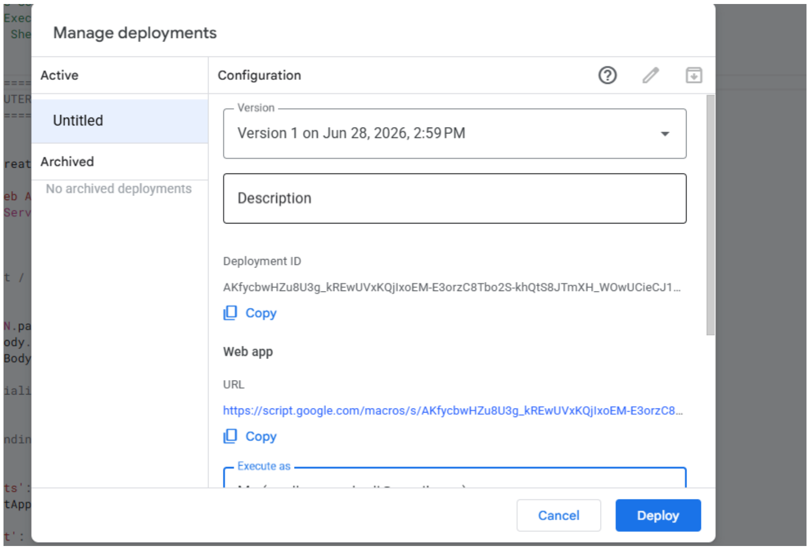
- **Step 1** – Create a new Google Spreadsheet.
- **Step 2** – Copy the Spreadsheet ID.
- **Step 3** – Create sheets such as:
  - Users
  - Appointments
  - Medicines
  - Health Records
  - AI Summaries
- **Step 4** – Update the Spreadsheet ID in Code.gs.

Activity 1.3: Deploy Backend

The backend was deployed as a Google Apps Script Web App.

Steps

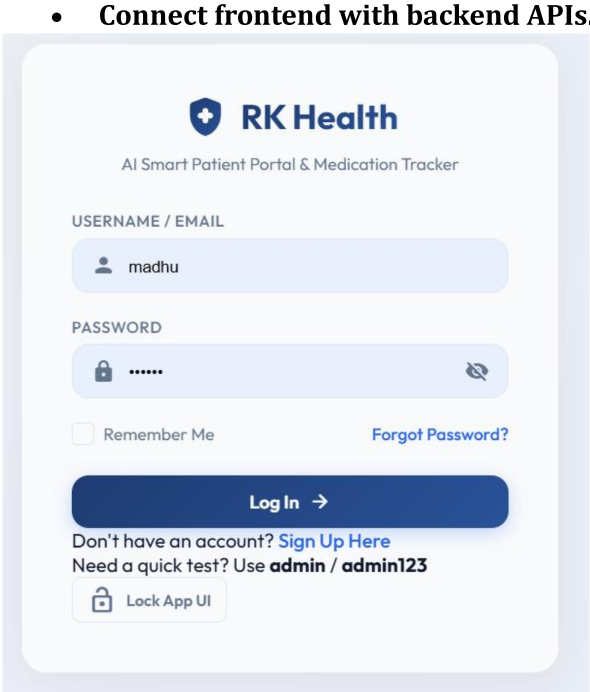
- Deploy → New Deployment.
- Select **Web App**.
- Execute as **Me**.
- Access: **Anyone**.
- Copy the generated Web App URL.



Milestone 2: Frontend Development and Core Modules
This milestone focused on designing the user interface and implementing the core healthcare management modules.

Activity 2.1: Develop Login & Dashboard
Steps

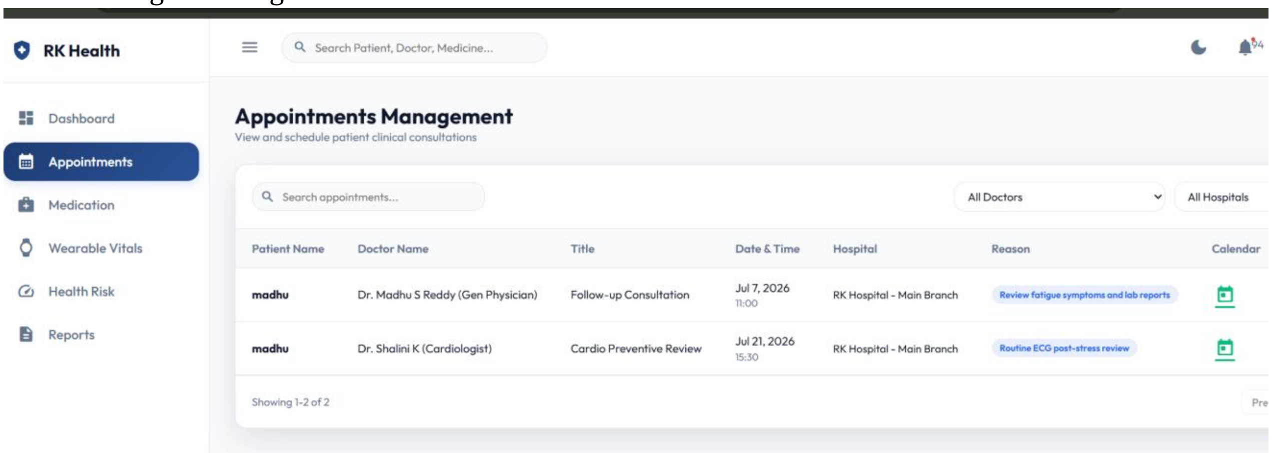
- **Create Login Page.**
- **Design Dashboard.**
- **Implement responsive UI.**
- **Connect frontend with backend APIs.**



Activity 2.2: Appointment Management Module

Steps

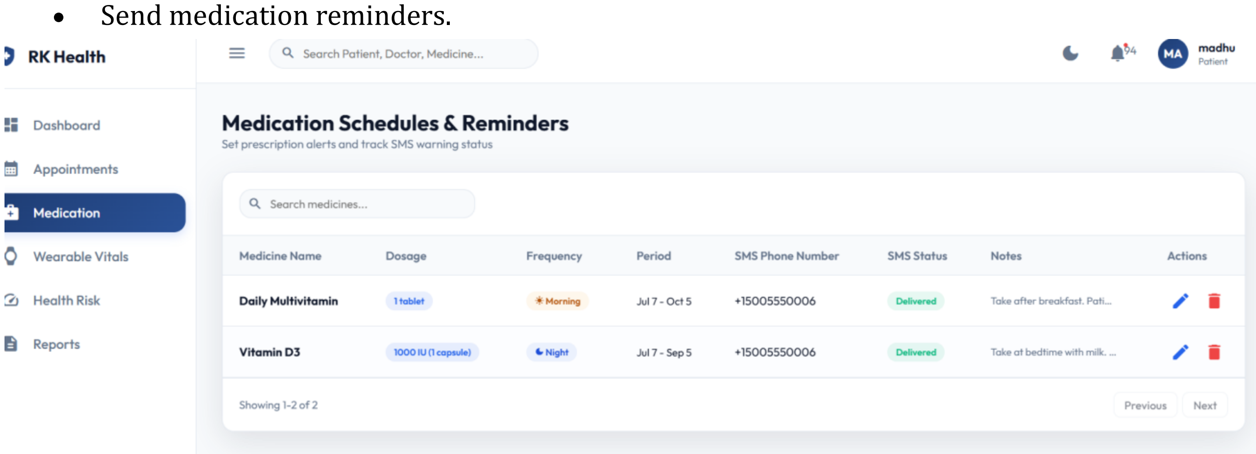
- Design Appointment Form.
- Create Appointment API.
- Save data to Google Sheets.
- Integrate Google Calendar.



Activity 2.3: Medication Reminder Module

Steps

- Create Medicine Form.
- Store medicines in Google Sheets.
- Integrate Twilio SMS.
- Send medication reminders.



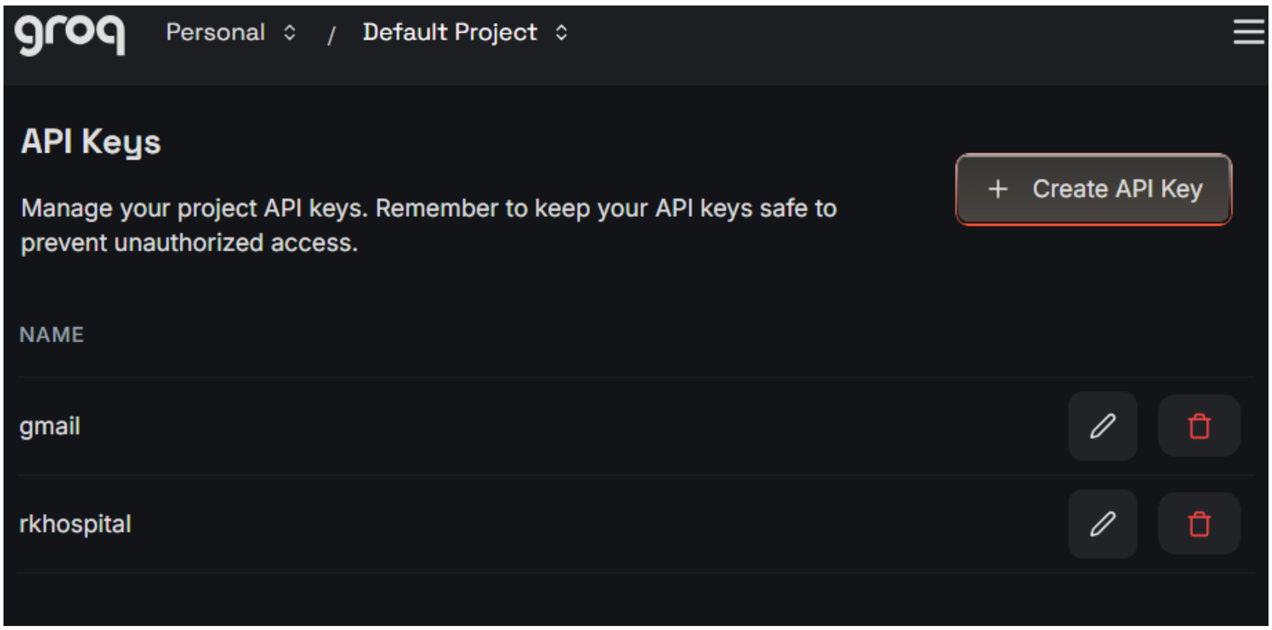
Milestone 3: AI Integration and Advanced Features

This milestone focused on integrating Artificial Intelligence and additional healthcare services.

Activity 3.1: Groq AI Integration

Steps

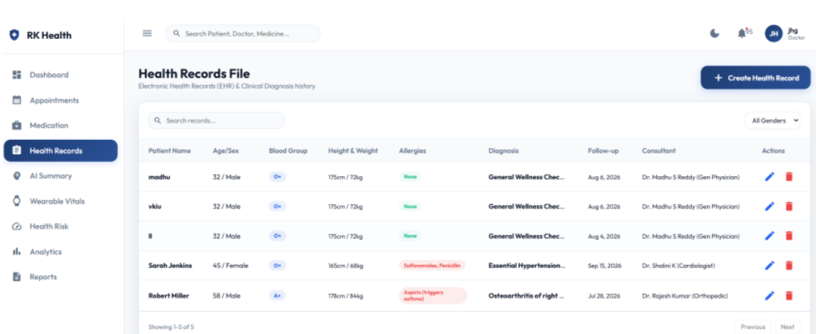
- Create Groq API Key.
- Configure Script Properties.
- Connect Groq API with Google Apps Script.
- Generate AI healthcare summaries.



Activity 3.2: Health Record Management

Steps

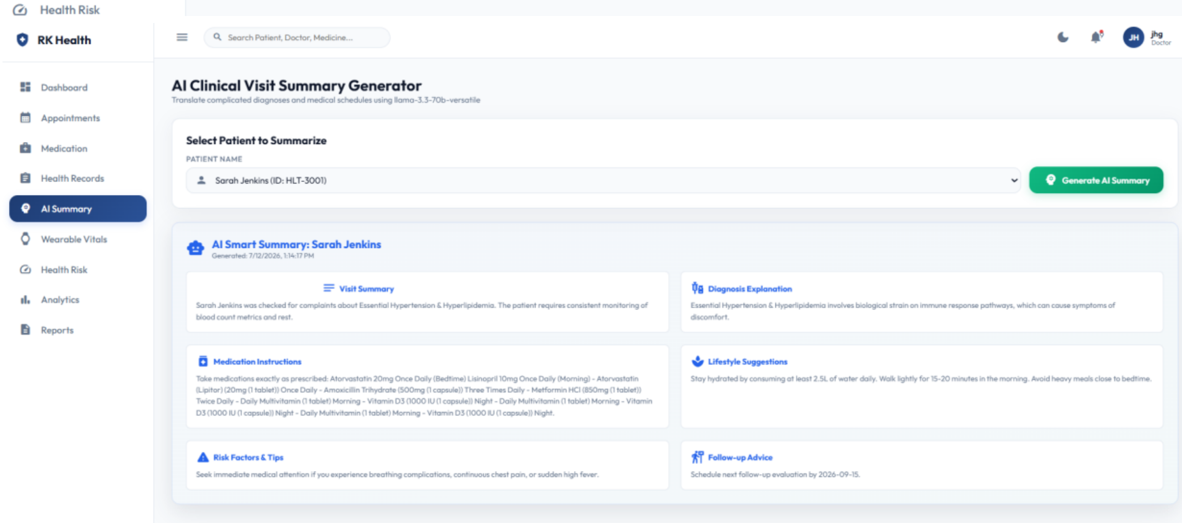
- Create Health Record Form.
- Store patient data.
- Display previous records.
- Enable report generation.



Activity 3.3: AI Summary Generation

Steps

- Collect patient visit details.
- Send data to Groq AI.
- Generate medical summary.
- Save summary in Google Sheets.



Milestone 4: Testing, Deployment and Documentation

This milestone focused on testing the application, fixing bugs, deploying the project, and preparing the documentation.

Activity 4.1: Functional Testing

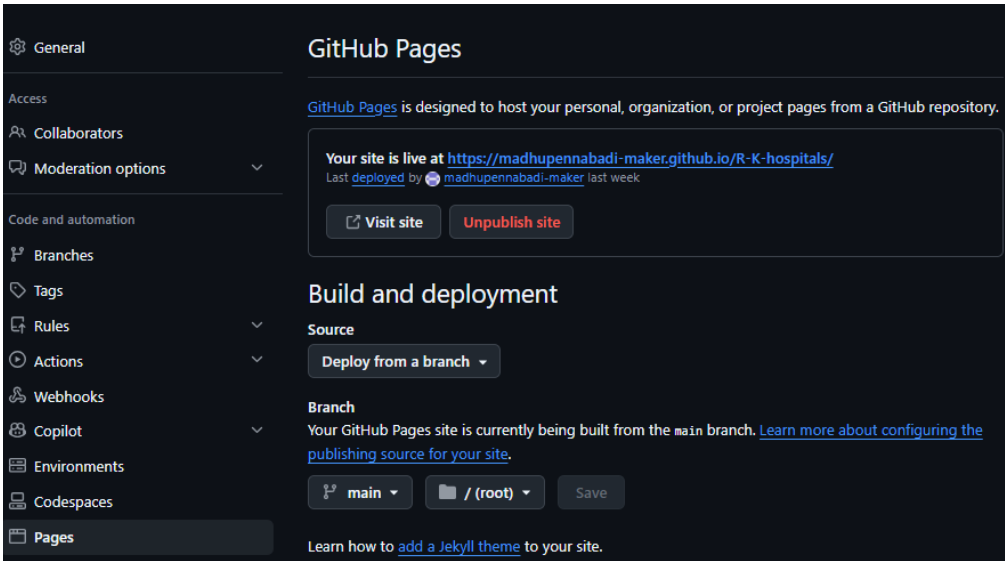
Steps

- Test Login.
- Test Appointment Module.
- Test Medicine Module.
- Test AI Summary.
- Verify Google Sheets data.

Activity 4.2: Deploy Application

Steps

- Upload source code to GitHub.
- Configure GitHub Pages.
- Publish website.
- Verify live deployment.



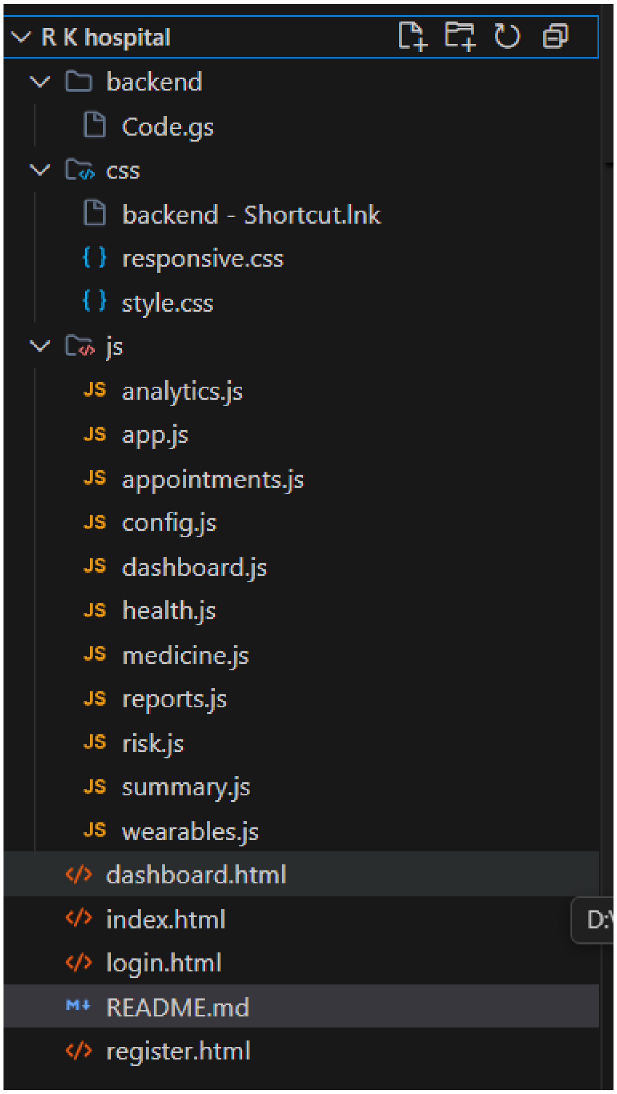
Activity 4.3: Documentation

Steps

- Prepare project report.
- Add architecture diagrams.
- Include screenshots.
- Prepare final presentation.

Project Folder Structure

The RK Health project follows a modular folder structure that separates the frontend, backend, styling, and application logic into different directories. This organization improves code readability, maintainability, and scalability while making it easier to manage different components of the application.



Activity: Groq AI Route Definition

Description:

The RK Health application includes an AI route for generating patient-friendly healthcare summaries. The frontend sends a POST request with the action "generateSummary" and the patient's visit details to the Google Apps Script backend. The backend validates the request, constructs a structured medical prompt, and securely communicates with the Groq AI API using the LLaMA 3.3-70B Versatile model. The generated summary is processed, stored in the Google Sheets database, and returned to the frontend as a JSON response.

Route Definition

POST / Google Apps Script Web App

Action:

generateSummary

Backend Routing Logic

```
case "generateSummary":
    responseData = generateSummary(payload);
    break;
```

Workflow Description

The RK Health system follows a structured workflow to provide efficient healthcare management services. Initially, the patient accesses the web application through the login page and authenticates using valid credentials. After successful authentication, the user is redirected to the dashboard, where various healthcare management services such as appointment scheduling, medication reminders, health record management, AI summary generation, and report generation are available.

When a patient performs an action, such as booking an appointment or adding medication details, the frontend sends the request to the **Google Apps Script backend**. The backend validates the request, processes the business logic, and stores or retrieves data from the **Google Sheets cloud database**. Depending on the requested service, the backend communicates with external APIs. Appointment details are integrated with **Google Calendar**, medication reminders are delivered through **Twilio SMS**, and patient visit information is processed by the **Groq AI API** to generate simplified healthcare summaries.

After processing is completed, the backend returns a JSON response to the frontend, where the updated information is displayed to the user. This workflow ensures secure data management, real-time updates, cloud accessibility, and an improved healthcare experience through intelligent automation and AI-powered assistance.

Workflow Steps

1. Patient opens the RK Health application.
2. User logs in to access the dashboard.
3. User selects a healthcare service (Appointments, Medicines, Health Records, AI Summary, or Reports).
4. Frontend sends the request to the Google Apps Script backend.

5. Backend validates and processes the request.
6. Data is stored or retrieved from Google Sheets.
7. External APIs (Groq AI, Google Calendar, Twilio SMS) are invoked when required.
8. Backend sends the processed response to the frontend.
9. Updated information is displayed to the patient.

Workflow

```
Frontend
      |
      ▼
Google Apps Script (doPost)
      |
      ▼
Action = "generateSummary"
      |
      ▼
generateSummary()
      |
      ▼
Groq AI API
      |
      ▼
AI Summary Generated
      |
      ▼
Google Sheets Database
      |
      ▼
Frontend Response
```

```
                 Patient
                    |
                    ▼
           Login / Authentication
                    |
                    ▼
              RK Health Dashboard
                    |
       ┌───────────┬───────────┬──────────────┐
       ▼           ▼           ▼              ▼
Appointments Medicines Health Records AI Summary
       |           |           |              |
       └───────────┴───────────┴──────────────┘
                    |
                    ▼
         Google Apps Script Backend
                    |
       ┌───────────┬─────────────┬─────────────┐
       ▼           ▼             ▼
Google Sheets Groq AI   Google Calendar
  Database       API          API
                    |
                    ▼
               Twilio SMS API
                    |
                    ▼
            Response to Frontend
                    |
                    ▼
           Updated Dashboard Display
```

Backend Deployment Process (Google Apps Script)

Step 1: Create Google Apps Script Project

- Open **Google Apps Script**.
- Create a new Apps Script project.
- Copy the Code.gs file into the editor.
- Save the project.

Step 2: Create Google Sheets Database

- Create a new Google Spreadsheet.
- Add the required sheets:

    - Appointments
    - Medicines
    - HealthRecords
    - AISummaries
- Copy the Spreadsheet ID.

Step 3: Configure Script Properties

Add the required API credentials:

- GROQ_API_KEY
- TWILIO_ACCOUNT_SID
- TWILIO_AUTH_TOKEN
- TWILIO_NUMBER

Step 4: Deploy Backend

- Click **Deploy → New Deployment**.
- Select **Web App**.
- Execute as **Me**.
- Who has access: **Anyone**.
- Click **Deploy**.
- Copy the generated Web App URL.
- Example:
- `https://script.google.com/macros/s/AKfycbxxxxxxxxxxxxxxxxxxxxxxxxx/exec`

1. GitHub Repository

Repository Link:

https://github.com/madhupennabadi-maker/R-K-hospitals

Description:

The GitHub repository contains the complete source code of the RK Health project, including the frontend developed with HTML, CSS, and JavaScript, and the backend implemented using Google Apps Script. It also maintains version control, project updates, and documentation, making the application easy to manage, maintain, and extend.

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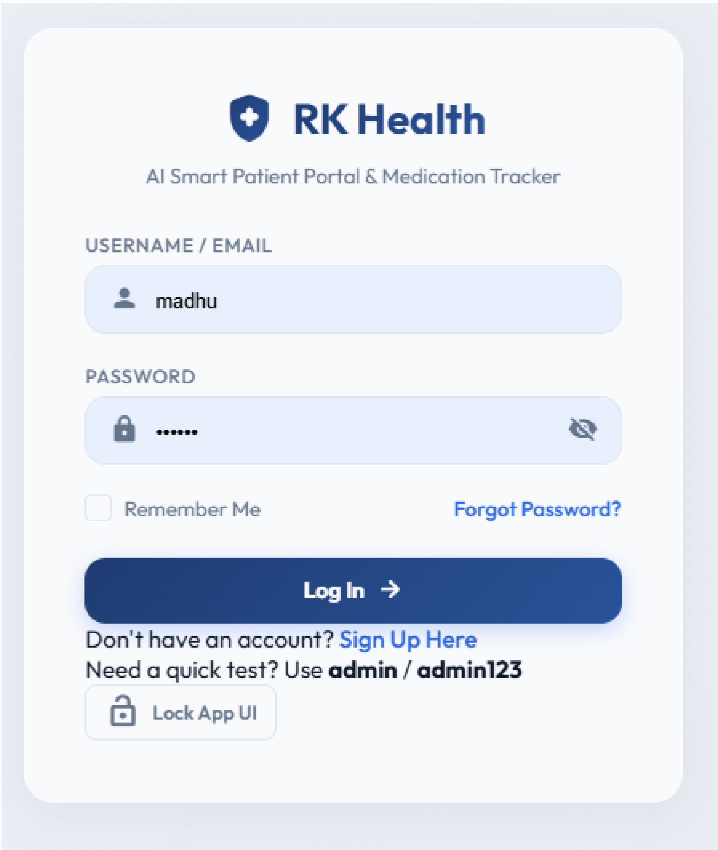
Graphical User Interface (GUI) of RK Health:

1. Login Page

Description:

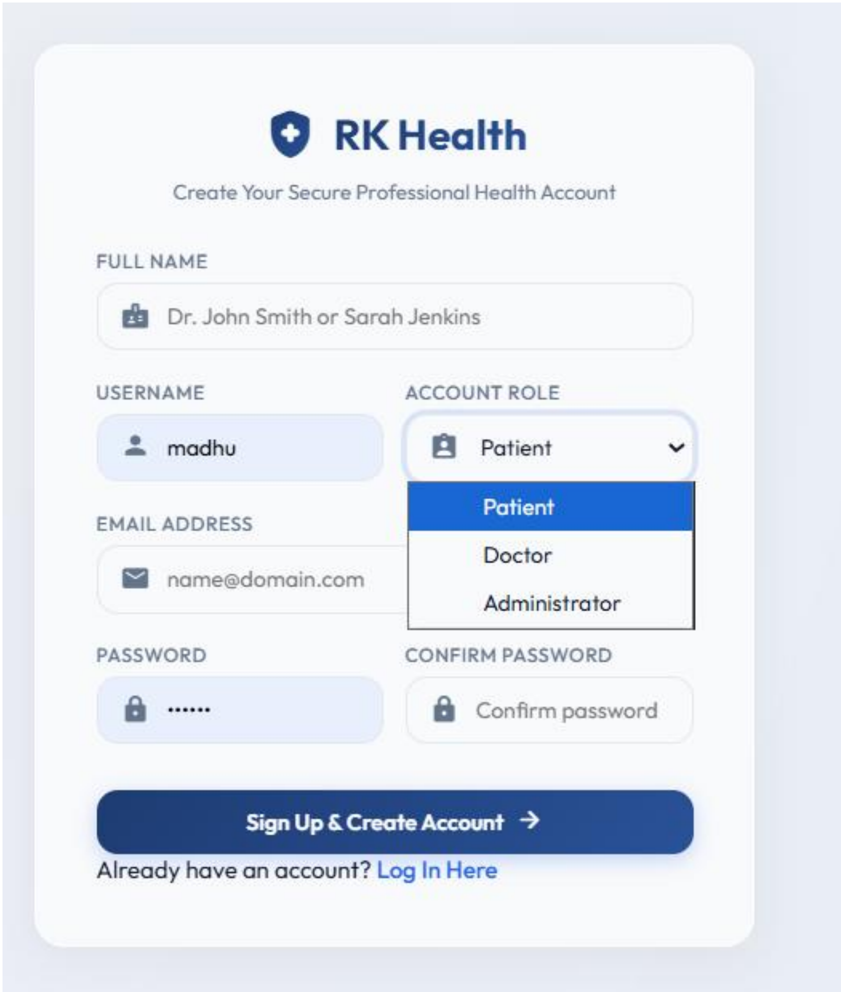
The Login Page provides secure access to the RK Health system by allowing registered users to log in using their credentials. It verifies user information and redirects authenticated users to their respective dashboards based on their roles.



2. Sign Up Page

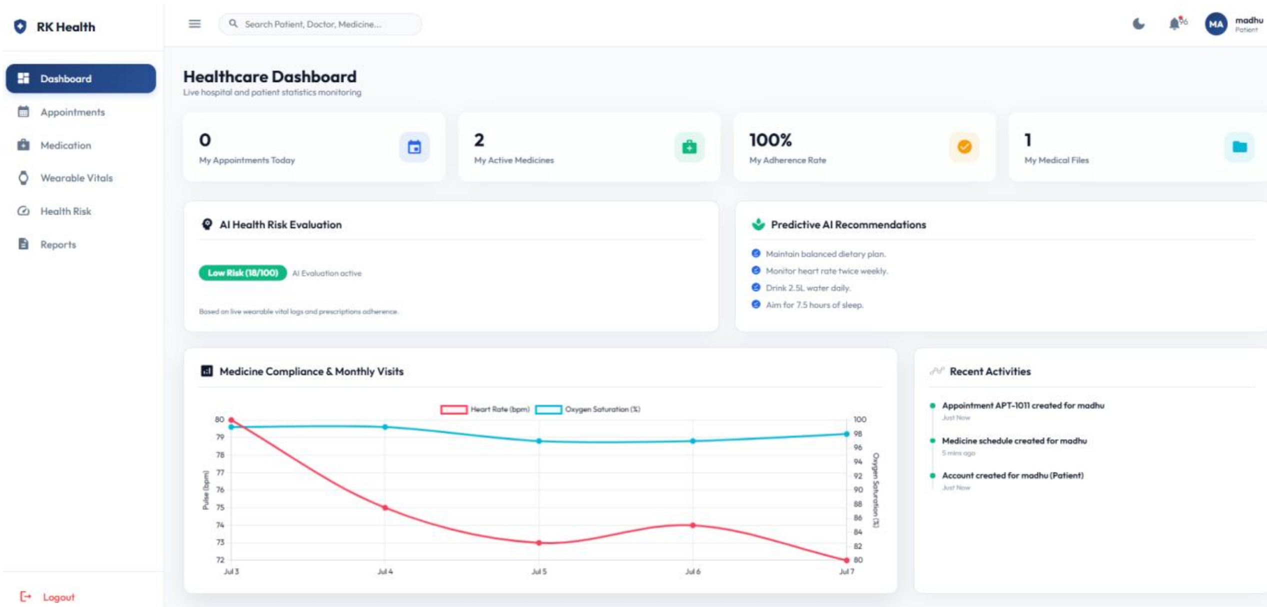
Description 1

The Sign Up Page allows new users to register for the RK Health system by providing their personal information, login credentials, and selecting an account role (Patient, Doctor, or Administrator). The entered details are securely stored in the cloud database, enabling role-based authentication and access to the application's features.



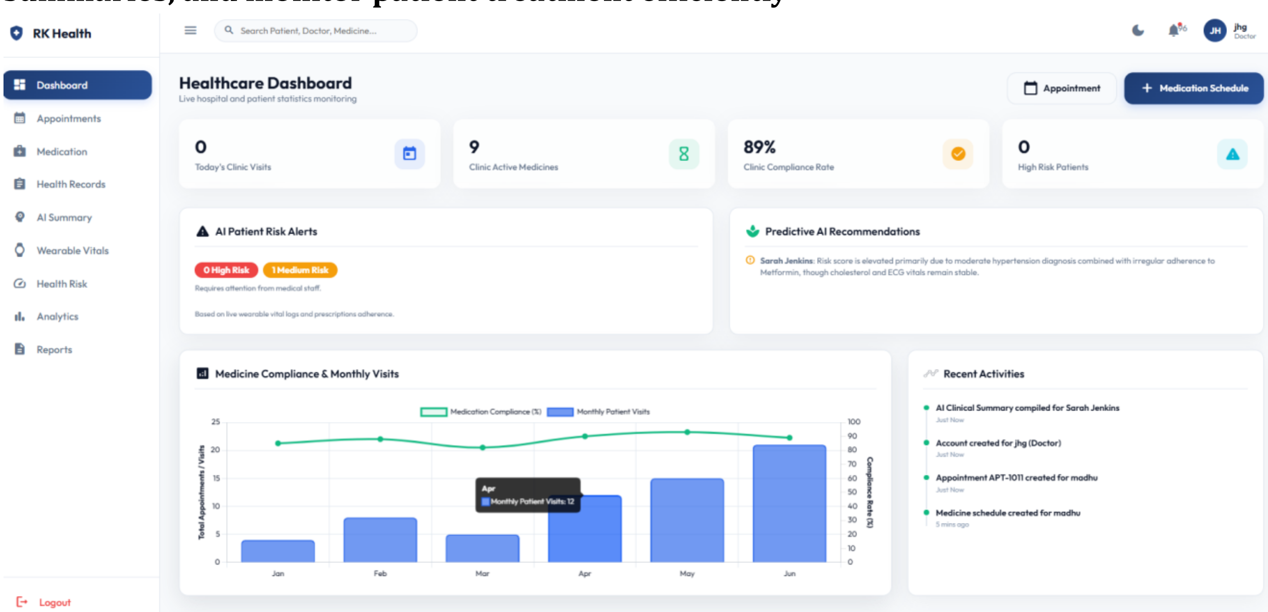
3. Patient Dashboard

The Patient Dashboard provides a centralized interface for patients to monitor their healthcare activities. It displays appointment schedules, active medications, AI-based health risk evaluation, predictive healthcare recommendations, wearable vital statistics, medical reports, and recent activities, enabling patients to efficiently manage their health through a single dashboard.



4. Doctor Dashboard

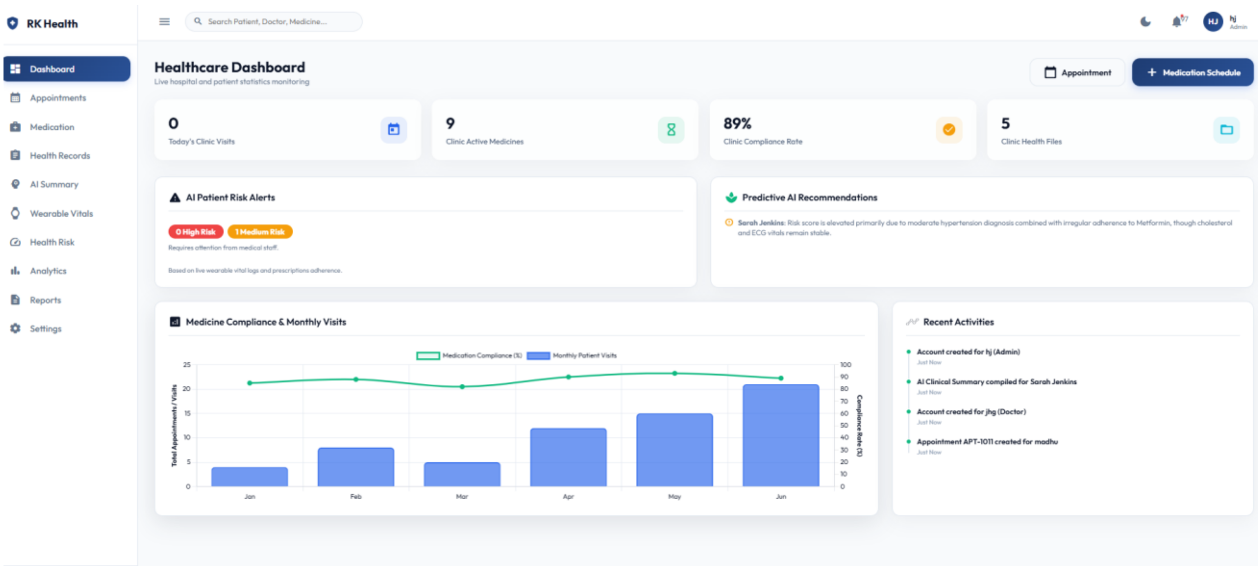
The Doctor Dashboard allows healthcare professionals to view patient information, manage appointments, update health records, review AI-generated summaries, and monitor patient treatment efficiently



5. Administrator Dashboard

Description:

The Administrator Dashboard provides complete control over the RK Health system, enabling administrators to manage users, monitor appointments, maintain healthcare records, oversee system activities, and ensure smooth application operation.


RK Health
Search Patient, Doctor, Medicine...
Dashboard
Appointments
Medication
Health Records
AI Summary
Wearable Vitals
Health Risk
Analytics
Reports
Settings
Healthcare Dashboard
Live hospital and patient statistics monitoring
Appointment
Medication Schedule
0
Today's Clinic Visits
9
Clinic Active Medicines
89%
Clinic Compliance Rate
5
Clinic Health Files
AI Patient Risk Alerts
0 High Risk
1 Medium Risk
Requires attention from medical staff.
Based on live wearable vital logs and prescriptions adherence.
Predictive AI Recommendations
Sarah Jenkins: Risk score is elevated primarily due to moderate hypertension diagnosis combined with irregular adherence to Metformin, though cholesterol and ECG vitals remain stable.
Medicine Compliance & Monthly Visits
Medication Compliance (%)
Monthly Patient Visits
Total Appointments / Visits
Compliance Rate (%)
Jan
Feb
Mar
Apr
May
Jun
Recent Activities
Account created for hj (Admin)
Just Now
AI Clinical Summary compiled for Sarah Jenkins
Just Now
Account created for jhg (Doctor)
Just Now
Appointment APT-1011 created for madhu
Just Now